George Zhang

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Canadian & Hong Kong citizen · eligible for U.S. internships (J-1); TN-eligible post-graduation

Education

University of Waterloo — Waterloo, ON

Bachelor of Computer Science (Honours, Co-op) — Artificial Intelligence Specialization — *Sep 2024 – May 2029*

- Relevant coursework: Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Optimization, Logic & Computation, OOP (C++).

Experience

Ford Motor Company — Waterloo, ON

Software Developer Intern — IVI Phone Apps — *May 2026 – Aug 2026*

- Spearheaded two automated workflows for the team's LLM log-analysis playbook — tag-differential/chunking updates, and a direct Confluence sync — cutting manual maintenance from the process.
- Shipped features across three in-vehicle apps (Dialer, Settings, Messenger) on Android Automotive OS, Ford's infotainment platform, in a multi-module Java/Kotlin codebase; closed all 64 issues at a 100% completion rate.

University of British Columbia — Vancouver, BC

Undergraduate Research Assistant — Centre for Brain Health — *Jun 2025 – Aug 2025*

- Led development of two full-stack hand-rehabilitation apps within PIKA, an AI Parkinson's-care platform deployed at Vancouver Coastal Health.
- Built a real-time computer-vision pipeline (Python, OpenCV, MediaPipe) detecting 7 hand-exercise gestures at ~90% accuracy with under 100 ms latency, fully on-device to meet PIPEDA/GDPR requirements.
- Designed activity-heatmap dashboards that cut clinician monitoring overhead by 30%; work presented at the 11th Singapore International Parkinson Disease & Movement Disorders Symposium.

Projects

Code-Aware RAG System | Python, Transformers — github.com/TheYellowDuck/RAG-codebase

- Engineered a retrieval-augmented-generation system answering codebase questions with citation-backed answers — AST-boundary chunking (tree-sitter), dense + BM25 fusion (RRF), a code graph with personalized PageRank, a validated listwise LLM reranker, and a RAGAS-style faithfulness check; provider-agnostic (Anthropic/OpenAI/local).
- Validated every choice as a measured ablation (recall@k, MRR, NDCG, bootstrap CIs, paired significance): the LLM reranker significantly improved ranking (MRR/NDCG, $p < 0.001$) while a cross-encoder proved net-negative on code.

RLHF Pipeline | PyTorch — github.com/TheYellowDuck/rlhf-pipeline

- Reproduced the LLM post-training stack (SFT, reward modeling, PPO) with the RL core written from scratch: GAE, the clipped policy-gradient surrogate, per-token KL-to-reference shaping, and an adaptive KL controller.
- Added DPO and GRPO alternatives, LoRA parameter-efficient fine-tuning, multi-GPU training (HuggingFace Accelerate), and an independent LLM-as-judge win-rate evaluation with position-bias control.

Autonomous LLM Web Agent | Python, Playwright — github.com/TheYellowDuck/web-agent

- Designed a ReAct + reflection agent (browser accessibility tree, provider-agnostic across Claude/GPT/Gemini) with a WebArena / Online-Mind2Web eval harness; showed the same agent scores 0.43 vs 0.61 by scorer alone, isolating measurement from real capability.

Technical Skills

Languages	Python, C++, Java, Kotlin, TypeScript/JavaScript, SQL
ML / AI	PyTorch, Transformers, RAG, RLHF (PPO/DPO/GRPO), LLM agents, prompt engineering, LLM-as-judge eval
Data	NumPy, Pandas, scikit-learn, Matplotlib, OpenCV, MediaPipe, embeddings, statistics
Tools	Git, Linux, Docker, GitHub Actions, HuggingFace, Playwright, Anthropic/OpenAI APIs, Confluence
Concepts	Data Structures & Algorithms, Evaluation & Statistical Rigor, Computer Vision, OOP, Workflow Automation

Awards & Competitive Programming

- President's Scholarship of Distinction & Entrance Scholarship in Mathematics, University of Waterloo
- ICPC Local (Waterloo) — Rank 28 / ~100 · 470+ LeetCode & 220+ DMOJ solved (1,497 points)
- Canadian Computing Competition — Junior: perfect score, Senior: top quartile · AIME Qualifier · COMC Distinction
- RoboCup International Junior Maze Rescue 2023 (Captain) — 2nd Place Supergroup, 17th Individual